Chapter 59 — Offline Media Downloads (OTT)

> **Part VII — Media & Files** · Complexity: **Complex** (≥40 interview questions)
>
> **Prerequisites:** Ch 58 (Media3 playback), Ch 18 (download manager), Ch 30 (WorkManager), Ch 46 (foreground services), Ch 3 (SSOT), Ch 25 (secure storage), Ch 19 (auth). Pairs with Ch 57 (posters), Ch 61 (large files), Ch 32 (sync).
>
> **Reference surface in StreamCart:** the "Download for offline" feature of the streaming side — letting users download movies/shows/podcasts (adaptive, DRM-protected) to watch without a network, with correct licensing, storage management, and playback that seamlessly switches online ↔ offline.

59.1 Problem Definition and Real-World Context

"Download to watch offline" (OTT: over-the-top video) is far harder than downloading a file (Ch 18). Streaming content is **adaptive** (HLS/DASH — many segments, multiple bitrates), usually **DRM-protected** (Widevine), **large** (gigabytes), **licensed** (time-limited, device-bound, concurrency-limited by the rights holder), and must **play back through the same player** (Media3, Ch 58) whether online or offline. A naive "save the video file" approach fails on every axis.

Concretely, teams hit:

- **Adaptive streams aren't a single file.** An HLS/DASH title is hundreds of segments across several bitrates. You must download a **chosen track/quality** (not all), store the segments, and let the player read them offline — Media3's **download modules** (`DownloadManager`, `DownloadService`) handle this; hand-rolling segment download is a nightmare.
- **DRM & offline licenses.** Protected content needs an **offline Widevine license** acquired at download time, stored securely, with an **expiry** and **renewal** — and the license (not just the bytes) governs playback. Deleting the download must **release the license**; an expired license must block playback and prompt renewal.
- **Huge storage, user choice, cleanup.** Downloads are gigabytes; users pick quality (storage vs fidelity), downloads must be **pausable/resumable** (Ch 18), survive reboot/process death, and be **cleaned up** (auto-delete watched, expiry, storage pressure). External/SD-card storage choices (Ch 25/62) matter.
- **Background, long-running, constrained.** Multi-GB downloads run in the **background** via a **foreground service / WorkManager** (Ch 46/30), respecting Wi-Fi-only/charging constraints (Ch 47) and resuming after interruptions.
- **Online ↔ offline playback switch.** The player (Ch 58) must transparently use the **downloaded copy** when offline (or to save data) and the **stream** otherwise — one playback experience.
- **Licensing policy enforcement.** Download limits (how many titles, how many devices), geo/entitlement rules, and expiry are **server-governed** (Ch 19/23); the client enforces the UX but the server authorizes.

The architecture: use **Media3's download stack** (`DownloadManager` + a `DownloadService`, a foreground service, Ch 46) to download a **selected track** of an adaptive stream with pause/resume/constraints (Ch 18/30/47); acquire and store an **offline DRM license** (Widevine) at download time, track its **expiry/renewal**, and **release** it on delete;

model downloads in the **SSOT** (Room, Ch 3) with state (queued/downloading/paused/completed/failed/expired) and progress, stored in **app-internal or user-chosen storage** (Ch 25/62); run downloads in the **background** (FGS/WorkManager, Ch 46/30) under constraints; make the **player** (Ch 58) transparently prefer the offline copy; **enforce licensing** (server-authorized limits/expiry, Ch 19/ 23); and **manage storage** (quality choice, cleanup, pressure). This chapter is about **licensed, DRM-protected, adaptive offline media done correctly**.

Three practical use cases

1. **Download a movie (StreamCart).** User taps Download, picks quality → Media3 downloads the selected track in the background (FGS, Wi-Fi-only), acquires an offline Widevine license, tracks progress in the SSOT, generates a poster (Ch 57); plays offline later.
2. **Expiry & renewal.** A downloaded title's license expires in 48h after first play → the UI shows the countdown; on expiry, playback is blocked and renewal is offered (server-authorized, Ch 19).
3. **Storage management.** Auto-delete watched downloads, honor a storage cap, let the user choose SD-card storage (Ch 62), and clean up on storage pressure.

Why this is architectural

Offline OTT combines **adaptive streaming + DRM licensing + huge background downloads + secure storage + player integration + server-enforced rights** — a system where the *license*, not just the bytes, is the asset, and correctness spans download, storage, expiry, playback, and entitlement. The decisions determine whether offline playback is legal, reliable, and storage-sane.

59.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Download a selected quality/track of an adaptive (HLS/DASH) title for offline playback.
F2	Acquire/store/renew/release an offline DRM (Widevine) license with expiry.
F3	Pause/resume/retry downloads; survive reboot/process death; background + constraints.
F4	Model downloads in the SSOT (state + progress); store securely (Ch 25/62).
F5	Play offline transparently via the same player (Ch 58); online ↔ offline switch.
F6	Enforce licensing policy (limits/expiry/entitlement) with server authorization (Ch 19/23).

Non-functional requirements

Attribute	Policy
Legality	DRM licenses respected; server-authorized limits/expiry/entitlement.
Reliability	Resumable; survives reboot/death; retries; completes multi-GB.
Storage	User-chosen quality/location; cleanup; pressure handling.
Battery/Data	Wi-Fi-only/charging constraints (Ch 47); background (Ch 46).
Security	Licenses + content stored securely (Ch 25); keys platform-held.
UX	Transparent offline playback; clear state/expiry/renewal.

Scope

- **In scope:** Media3 download stack (`DownloadManager` / `DownloadService` FGS, Ch 46), track/quality selection for download, offline DRM (Widevine) license acquire/store/ renew/release + expiry, pause/resume/retry + reboot/death survival + constraints (Ch 18/30/47), SSOT download model (Ch 3), secure/user-chosen storage (Ch 25/62), transparent offline playback via Media3 (Ch 58), server-enforced licensing (Ch 19/23), storage management/cleanup, poster generation (Ch 57).
- **Out of scope:** playback state detail (Ch 58), generic downloads (Ch 18), WorkManager/ Doze internals (Ch 30/47), storage APIs detail (Ch 62), sync (Ch 32), thumbnail generation (Ch 57).

Assumptions & constraints

- **Media3** download modules + `DrmSessionManager` (Widevine), `DownloadService` (FGS, Ch 46), Room/DataStore (Ch 3/4), WorkManager (Ch 30). `minSdk 24`. Content is adaptive + DRM-protected + licensed; the **license governs playback** (bytes alone are useless); the **server authorizes** downloads/renewals/limits.

59.3 System Design Fundamentals

- **The license is the asset, not the bytes.** For DRM content, downloading the (encrypted) media segments is necessary but useless without a valid **offline license**. The license is **acquired** at download time (from the license server, authed, Ch 19), **stored** securely, has an **expiry** (and possibly a "playback window" that starts on first play), can be **renewed**, and must be **released** when the download is deleted (so the rights holder's device/concurrency accounting is correct). **Playback is gated by the license**, not file presence.
- **Use Media3's download stack (don't hand-roll).** Media3 provides:
  - **`DownloadManager`:** manages a queue of downloads, pause/resume, retries, progress, persistence (it has its own index), and constraint handling — purpose-built for adaptive media.
  - **`DownloadService`:** a **foreground service** (Ch 46, type `dataSync`) that runs the downloads in the background with a notification and survives app death/reboot.
  - **`DownloadHelper`:** selects which **tracks/qualities** to download from an adaptive manifest (so you download one bitrate + chosen audio/subtitle, not everything).

    - **Offline DRM:** `OfflineLicenseHelper` / `DefaultDrmSessionManager` acquire and persist the offline license; playback uses it. Hand-rolling segment download + DRM is a massive, error-prone effort — use the stack.
- **Track/quality selection (storage vs fidelity).** From the adaptive manifest, let the user (or a policy) choose a **bitrate/resolution** and which **audio/subtitle** tracks to download — this determines the storage footprint. You download a **single rendition** (plus chosen extra tracks), not all bitrates.
- **Background, long-running, constrained (Ch 46/30/47).** Multi-GB downloads run in the `DownloadService` (FGS) with **constraints** (Wi-Fi-only, charging, battery-not-low, Ch 47), **pause/resume** (user or constraint-driven), **retry with backoff** (Ch 2), and **survival** across reboot/process death (DownloadManager persists its index; a `BOOT_COMPLETED` /service restart resumes).
- **SSOT model (Ch 3).** Mirror download state into **Room** for the UI (Ch 6): `Download{id, title, quality, state, progress, bytesDownloaded, totalBytes, licenseExpiry, storagePath, account}` with states **queued / downloading / paused / completed / failed / expired**. Media3's DownloadManager is the engine; the SSOT is what the app/UI observes and what survives.
- **Secure, user-chosen storage (Ch 25/62).** Store downloaded (encrypted) media in **app-specific storage** (internal or app-specific external/SD — large content often goes to SD, Ch 62), not shared storage. The content is DRM-encrypted anyway; keep licenses in secure storage (Ch 25). Honor the user's **storage location** choice and a **storage cap**.
- **Transparent offline playback (Ch 58).** The **same Media3 player** plays the downloaded copy: when offline (or configured to prefer downloads), the player's `MediaSource` reads from the **download cache** and uses the **offline license**; when online and not downloaded, it streams. The user sees **one playback experience**; the app chooses the source based on availability/preference.
- **Expiry, renewal, release:**
    - **Expiry:** the offline license has a hard expiry and/or a playback window; the UI surfaces the countdown; on expiry, **block playback** and offer **renewal** (re-acquire from the server, authed, if still entitled).
    - **Release:** deleting a download must **release the offline license** (so the rights holder's accounting is correct) and remove the media + SSOT row.
- **Server-enforced licensing (Ch 19/23).** Download **limits** (titles/devices), **entitlement** (is the user allowed to download this?), **geo**, and **expiry** are decided **server-side**; the license server issues/renews offline licenses only to entitled, within-limit requests. The client enforces the **UX** (show limits, disable download) but the **server authorizes** — a modified client can't grant itself an offline license without the server.
- **Cleanup & storage pressure.** Auto-delete watched downloads (policy), delete on expiry, honor the cap, and react to **storage pressure** (stop/prompt). Orphaned media/licenses must be reclaimable.

Core insight: **offline OTT is about the license as much as the bytes — use Media3's download stack (`DownloadManager` + a `DownloadService` foreground service + `DownloadHelper` for track selection) to download a chosen rendition of an adaptive stream in the background under constraints (Ch 46/30/47), acquire/store/renew/release an offline Widevine license (expiry-governed, server-authorized, Ch 19/23), model downloads in the SSOT (Ch 3) in secure/user-chosen storage (Ch 25/62), and let the same Media3 player (Ch 58) play the downloaded copy transparently offline — with the server enforcing entitlement, limits, and expiry because the license, not file presence, authorizes playback.**

59.4 High-Level Architecture


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Components & ownership

- **Server** authorizes entitlement/limits and issues/renews offline licenses (Ch 19/23).
- **Media3 download stack** (`DownloadHelper` → `DownloadManager` → `DownloadService` FGS) downloads a chosen rendition in the background.
- **Offline license** (Widevine) is acquired, stored securely, expiry-tracked, released on delete.
- **SSOT** mirrors state/progress; the **player** (Ch 58) plays the offline copy transparently.

Direction: entitlement → track select → background download + license acquire → SSOT/storage → transparent offline playback; delete releases license.

59.5 Detailed Component Design


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Component	Responsibility
`DownloadsRepository`	app-facing API; mirror Media3 DownloadManager into SSOT (Ch 3)
`DownloadEntity`	SSOT row (state/progress/expiry/path/account)
`OfflineLicenseController`	acquire/renew/release/validate offline Widevine license
`DownloadServiceImpl`	Media3 `DownloadService` (FGS, background, Ch 46)

59.6 End-to-End Data Flow

Downloading a DRM title


Downloading a DRM title

Transparent offline playback + expiry


Transparent offline playback + expiry

59.7 Complete Kotlin Implementation

Package: `com.streamcart.offline`.

59.7.1 Media3 DownloadService (FGS, Ch 46) + DownloadManager

```
@AndroidEntryPoint
class StreamCartDownloadService : DownloadService(
    FOREGROUND_NOTIFICATION_ID, DEFAULT_FOREGROUND_NOTIFICATION_UPDATE_INTERVAL,
    DOWNLOAD_CHANNEL_ID, R.string.downloads_channel, 0,   // notification (Ch 50)
) {
    @Inject lateinit var downloadManager: DownloadManager   // Media3, provided singleton
    override fun getDownloadManager() = downloadManager
    override fun getScheduler() = WorkManagerScheduler(this, "downloads")   // resume via
WorkManager (Ch 30)
    override fun getForegroundNotification(downloads: List<Download>, notMetRequirements: Int) =
        downloadNotificationHelper.buildProgressNotification(this, R.drawable.ic_download, null,
            /* message */ null, downloads, notMetRequirements)   // progress (Ch 46/50)
}

// DownloadManager provided with constraints (Wi-Fi-only etc., Ch 47) + cache in chosen storage
(Ch 25/62).
@Provides @Singleton
fun downloadManager(@ApplicationContext ctx: Context, db: StandaloneDatabaseProvider,
                    cache: Cache, dataSourceFactory: HttpDataSource.Factory): DownloadManager =
    DownloadManager(ctx, db, cache, dataSourceFactory, Executors.newFixedThreadPool(4)).apply {
        requirements = Requirements(Requirements.NETWORK_UNMETERED)   // Wi-Fi-only (Ch 47)
        maxParallelDownloads = 2
    }
```

59.7.2 Track/quality selection via DownloadHelper

```
suspend fun buildDownloadRequest(mediaItem: MediaItem, maxHeight: Int): DownloadRequest {
    val helper = DownloadHelper.forMediaItem(context, mediaItem, renderersFactory,
dataSourceFactory)
    helper.prepare(object : DownloadHelper.Callback {
        override fun onPrepared(h: DownloadHelper) {
            // Select ONE rendition (chosen quality) + desired audio/subtitle -> not all
bitrates
            val params =
DefaultTrackSelector.Parameters.Builder(context).setMaxVideoSize(Int.MAX_VALUE,
maxHeight).build()
            for (period in 0 until h.periodCount) {
                h.clearTrackSelections(period); h.addTrackSelection(period, params)
            }
        }
        override fun onPrepareError(h: DownloadHelper, e: IOException) { /* surface error */ }
    })
    return helper.getDownloadRequest(/* id */ mediaItem.mediaId, /* data */ null)
}
// DownloadService.sendAddDownload(context, StreamCartDownloadService::class.java, request,
/*foreground*/ false)
```

59.7.3 Offline DRM license: acquire, renew, release (Widevine)

```
@Singleton
class OfflineLicenseController @Inject constructor(
    private val httpFactory: HttpDataSource.Factory,      // authed (Ch 19), pinned (Ch 23)
    private val licenseStore: SecureLicenseStore,         // Ch 25
) {
    private fun helper() = OfflineLicenseHelper.newWidevineInstance(LICENSE_URL, httpFactory,
DrmSessionEventListener.EventDispatcher())

    /** At download time: server authorizes, then acquire + store the offline key set. */
    suspend fun acquire(mediaId: String, drmInitData: DrmInitData): ByteArray =
withContext(Dispatchers.IO) {
        val keySetId = helper().downloadLicense(drmInitData)   // offline Widevine license
        licenseStore.put(mediaId, keySetId)                    // store securely (Ch 25)
        keySetId
    }
    suspend fun renew(mediaId: String, drmInitData: DrmInitData) = acquire(mediaId, drmInitData)
// re-acquire if entitled
    suspend fun release(mediaId: String) = withContext(Dispatchers.IO) {
        licenseStore.get(mediaId)?.let { helper().releaseLicense(it) }   // release at rights
server
        licenseStore.remove(mediaId)                                     // delete local
    }
    fun remainingMs(mediaId: String): Long =
        licenseStore.get(mediaId)?.let { helper().getLicenseDurationRemainingSec(it).first *
1000L } ?: 0L
}
```

59.7.4 SSOT model + repository (mirror Media3 into Room, Ch 3)

```
@Entity(tableName = "downloads")
data class DownloadEntity(
    @PrimaryKey val id: String, val title: String, val quality: String, val state: String,  //
queued/downloading/paused/completed/failed/expired
    val progress: Float, val bytesDownloaded: Long, val totalBytes: Long,
    val licenseExpiryMs: Long, val storagePath: String, val account: String,
)

@Singleton
class DownloadsRepository @Inject constructor(
    private val dao: DownloadDao, private val license: OfflineLicenseController,
    @ApplicationContext private val ctx: Context, private val activeAccount:
ActiveAccountProvider,  // Ch 42
) {
    fun observe() = dao.observe(activeAccount.require().id)        // UI observes SSOT (Ch 6)

    suspend fun delete(id: String) {
        DownloadService.sendRemoveDownload(ctx, StreamCartDownloadService::class.java, id,
false)  // remove media
        license.release(id)                                          // RELEASE the license (rights
accounting)
        dao.delete(id)
    }
    // A DownloadManager.Listener updates the SSOT rows as Media3 progresses/completes/fails.
    fun bindListener(dm: DownloadManager) = dm.addListener(object : DownloadManager.Listener {
        override fun onDownloadChanged(dm: DownloadManager, d: Download, e: Exception?) {
            scope.launch { dao.upsert(d.toEntity(license.remainingMs(d.request.id),
activeAccount.require().id)) }
        }
    })
}
```

59.7.5 Transparent offline playback (Media3, Ch 58)

```
// The player prefers the downloaded copy + offline license when available/valid; else streams.
fun buildMediaSource(mediaId: String, streamUri: Uri, cache: Cache): MediaSource {
    val offlineKeySetId = licenseStore.get(mediaId)
    val dataSourceFactory = CacheDataSource.Factory().setCache(cache)          // reads the
download cache if present
        .setUpstreamDataSourceFactory(httpFactory)                             // else network
(Ch 58)
    val drmConfig = MediaItem.DrmConfiguration.Builder(C.WIDEVINE_UUID)
        .setKeySetId(offlineKeySetId)                                          // use OFFLINE
license if present
        .build()
    val item =
MediaItem.Builder().setMediaId(mediaId).setUri(streamUri).setDrmConfiguration(drmConfig).build()
    return DefaultMediaSourceFactory(dataSourceFactory).createMediaSource(item)
}
// On play: if license expired -> block + offer renewal (server authed).
```

59.7.6 Module structure

```
:feature:downloads  StreamCartDownloadService (FGS, Ch 46), DownloadsRepository,
DownloadEntity/Dao (Ch 3)
:core:drm           OfflineLicenseController, SecureLicenseStore (Ch 25)
:core:playback      offline-aware MediaSource for the player (Ch 58)
:core:storage       download cache location (app-specific/SD, Ch 62) + storage policy
# Entitlement/limits/license server -> backend (Ch 19/23). Posters -> Ch 57.
```

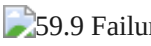
59.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Download + play same title	partial-file playback	play only when state=completed; Media3 cache handles partial for progressive, gate for DRM
License acquire fails mid-download	unplayable download	mark failed; retry license; don't mark completed without license
Delete during download	orphaned bytes/license	cancel download, release license, remove media atomically
Reboot/process death mid-download	stuck download	DownloadManager persists index; Scheduler/BOOT resumes (Ch 30)
Concurrent downloads exceed storage	failures	`maxParallelDownloads` + storage check before enqueue
Expiry during playback	illegal continued play	check license validity at play start + periodically; block on expiry
Account switch	cross-account downloads shown	filter SSOT by active account (Ch 42)

- **Completion means bytes AND license:** never mark a download "completed/playable" until **both** the media is fully downloaded **and** a valid offline license is stored — otherwise the user sees a download that won't play (common bug).
- **Delete must be atomic across three things:** cancel/remove the media, **release the license** (rights accounting), and delete the SSOT row — a partial delete leaves orphans or over-counts the user's device/title limit.
- **DownloadManager persistence + a Scheduler** make downloads survive reboot/process death: the index is persisted and a `WorkManagerScheduler` / `BOOT_COMPLETED` resumes under constraints (Ch 30/47).

59.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Downloaded but won't play	playback error	ensure license acquired before "completed"; renew	plays
License expired	validity check	block + offer renewal (server authed, Ch 19)	legal
Download interrupted	state	resume (DownloadManager + Scheduler, Ch 30)	completes
Reboot loses downloads	missing	persisted index + BOOT resume	survives
Storage full	enqueue/write fail	check before enqueue; cleanup; prompt (Ch 62)	handled
Delete leaves orphans	storage audit	atomic delete (media+license+row)	clean
Over device/title limit	server deny	enforce server-side; show limit UX	legal
Content plays after unsubscribe	entitlement	license expiry + server renewal check	enforced


59.9 Failure Scenarios and Recovery

> **The license governs everything.** Bytes on disk are worthless without a valid offline license, so "downloaded" means **media + license**, playback is **gated by license validity**, delete must **release the license**, and limits/expiry/entitlement are **server-authorized**. Treat the license as the real asset and most offline-OTT bugs (won't-play, illegal-play, over-limit, orphans) disappear.

59.10 Scaling the Implementation

- **Level 1.** Media3 `DownloadManager` + `DownloadService` (FGS) downloading a chosen track; SSOT model; basic playback from cache.
- **Level 2.** Offline Widevine license acquire/validate/release; expiry UI; pause/resume/ constraints (Ch 47); reboot/death resume (Ch 30).
- **Level 3 — production.** Server-enforced entitlement/limits/renewal (Ch 19/23); transparent online ↔ offline playback (Ch 58); storage choice (SD, Ch 62) + cleanup; posters (Ch 57); per-account (Ch 42).
- **Level 4 — large.** Quality/track selection UX + storage estimates; auto-delete-watched policies; download queue management; smart downloads (auto-download next episode); analytics (completion, expiry churn).
- **Level 5 — enterprise.** Rights-management governance (device/title/geo limits), multi- DRM, offline-license renewal SLAs, storage SLOs, cross-device download accounting, download-QoE dashboards.

Avoid overengineering: use Media3's download stack + offline license helper + SSOT; don't hand-roll segment download or DRM. Add quality UX/cleanup policies as needed.

59.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Download engine	Media3 download stack	custom segment downloader (Ch 18)	Media3 for adaptive+DRM; custom only for non-adaptive files
Playback auth	Offline license (gated)	file presence	License-gated always (DRM legality)
Track scope	Selected rendition	all bitrates	Selected (storage) — download one quality
Background	`DownloadService` (FGS)	WorkManager raw	DownloadService (purpose-built, resumable, Ch 46)
Storage	App-specific / SD (Ch 62)	shared storage	App-specific/SD (security + size); not shared
Limits/expiry	Server-authorized	client-only	Server (bypassable client)
Delete	Atomic media+license+row	remove bytes only	Atomic (release license/accounting)

Media3 download stack vs custom downloader (Ch 18): a generic resumable downloader (Ch 18) handles a single file, but **adaptive + DRM** media needs manifest parsing, track selection, multi-segment download, a persistent index, and offline-license acquisition — which Media3's `DownloadManager` / `DownloadService` / `DownloadHelper` / `OfflineLicenseHelper` provide and integrate with the

player (Ch 58). Hand-rolling this is a huge, fragile effort; use Media3's stack for OTT, and reserve the Ch 18 downloader for plain files.

License-gated vs file-presence playback: treating "the file is on disk" as permission to play ignores DRM — the content is encrypted and legally governed. Playback must be **gated by a valid offline license** (acquired at download, expiry-bound, server- authorized). This is both a legal requirement and the mechanism for expiry/limits; file presence alone must never authorize playback.

Selected rendition vs all bitrates: downloading every bitrate of an adaptive title wastes gigabytes. Download a **single chosen rendition** (quality the user picked) plus desired audio/subtitle tracks via `DownloadHelper` — the whole point of offline is a bounded, user-chosen footprint, not a full mirror of the stream.

59.12 Security & DRM Considerations

- **Offline license handling is the core security concern:** keys are managed by the platform DRM stack (never exposed to app code); the **offline key set id** is stored securely (Ch 25); the license has **expiry** and must be **released** on delete (rights accounting). Don't attempt to extract/persist raw keys.
- **Server authorizes entitlement/limits/renewal (Ch 19/23):** the license server issues/ renews offline licenses only to entitled, within-limit, correct-geo requests — a modified client can't self-grant. License + manifest requests go over the **authed, pinned** stack (Ch 19/23).
- **Secure output for protected playback (Ch 58):** `FLAG_SECURE`/secure surfaces for protected video to block capture.
- **Store content in app-specific storage (Ch 25/62):** downloaded (encrypted) media in app-specific internal/external dirs, not shared storage; licenses in secure storage.
- **Per-account isolation (Ch 42):** downloads and licenses are per-account; switching/ logout must not expose another account's downloads; release/clear appropriately.
- **No PII/keys/signed URLs in logs** (Ch 28).
- Client vs backend: the backend owns **entitlement, device/title limits, license issuance/renewal/expiry, and geo**; the client downloads within those, stores securely, gates playback on the license, and uses secure surfaces — it never bypasses DRM or self-authorizes.

59.13 Performance and Resource Optimization

- **Download one rendition** (chosen quality) — the main storage/bandwidth lever.
- **Constraints** (Wi-Fi-only, charging, Ch 47) avoid cellular/battery cost; schedule in good conditions.
- **Parallelism bounded** (`maxParallelDownloads`) to avoid IO/CPU thrash.
- **Resume, don't restart** (DownloadManager persistence) — never re-download on interruption.
- **Storage management:** quality-based size estimates, auto-delete-watched, cap, SD-card option (Ch 62), react to pressure.
- **Poster/metadata** generated once (Ch 57), cached.
- **Efficient SSOT updates:** debounce progress writes (Ch 31) — don't write every percent.
- **Profiling:** download completion rate/time, resume rate, storage used, license-acquire latency/failure, expiry-driven churn, playback-from-download success.

59.14 Testing Strategy

Media3 download components test with fakes; license/SSOT logic in isolation; instrumentation for real DRM/background.

```
class DownloadsRepositoryTest {
    @Test fun `completed requires media and license`() = runTest {
        // simulate media complete but license missing -> state != completed (or flagged)
    }
    @Test fun `delete releases license and removes row`() = runTest {
        repo.delete("m1")
        assertTrue(fakeLicense.released.contains("m1")); assertNull(fakeDao.get("m1"))  //
atomic
    }
    @Test fun `downloads filtered by active account`() = runTest { /* Ch 42 */ }
}

class OfflineLicenseControllerTest {
    @Test fun `expired license blocks playback`() { /* remainingMs <= 0 -> blocked */ }
    @Test fun `renew re-acquires when entitled`() { /* server ok -> new keySetId */ }
}
// Instrumentation: real DownloadService background download + reboot resume + offline playback
(test DRM/content).
```

What to test: completion requires media + license; delete is atomic (media + license + row); expiry blocks playback + renewal re-acquires; track selection downloads one rendition; resume after interruption/reboot (DownloadManager + Scheduler); constraints honored; per-account filtering; transparent offline playback picks the cache + offline license. Instrumentation with test DRM content for end-to-end.

59.15 Production Observability

- **Metrics:** download **completion rate/time**, resume rate, failure reasons; **license- acquire latency/failure**; **expiry-driven churn** (titles expiring unwatched); storage used per user; playback-from-download success; over-limit denials; orphaned media/ license count (0).
- **Logs:** `download{id, state, quality, bytes, license_expiry, account}` — no signed URLs/keys/PII (Ch 28).
- **Debugging:** downloaded-but-won't-play → license not acquired before "completed" / expired; stuck downloads → Scheduler/constraints/storage; orphans → non-atomic delete; illegal play → license not validated at play; over-limit → server not enforcing.
- **Alerting:** license-acquire failure spike (license server); completion-rate drop; storage-pressure failures.
- **Dashboards:** completion/resume/license-acquire funnel; expiry churn; storage usage.

59.16 System Design Interview Preparation

> ≥40 questions.

Fundamentals

1. *Why is offline OTT hard?* Adaptive + DRM + huge + licensed + player-integrated.
2. *The real asset?* The **offline license**, not the bytes.
3. *Download engine?* Media3 download stack (`DownloadManager` / `DownloadService` / `DownloadHelper`).
4. *Why not a generic downloader (Ch 18)?* Adaptive + DRM need manifest/track/segment/ license handling.
5. *Where does the download run?* `DownloadService` (FGS, Ch 46), background, resumable.

DRM/license

6. *Offline license?* Widevine license acquired at download, stored securely, expiry-bound.
7. *What gates playback?* A valid offline license (not file presence).
8. *Expiry?* Hard expiry / playback window; block + renew on expiry.
9. *Renewal?* Re-acquire from server if still entitled (Ch 19).
10. *Delete?* Release the license (rights accounting) + remove media + SSOT row (atomic).
11. *Keys in app code?* No — platform DRM stack holds them.

Download mechanics

12. *Track selection?* `DownloadHelper` — download one rendition + chosen audio/subtitle.
13. *Why one rendition?* Storage (not all bitrates).
14. *Constraints?* Wi-Fi-only/charging (Ch 47).
15. *Pause/resume?* DownloadManager supports; survives reboot/death via Scheduler (Ch 30).
16. *Parallelism?* `maxParallelDownloads`.

SSOT/storage

17. *State model?* Room SSOT: queued/downloading/paused/completed/failed/expired + progress (Ch 3).
18. *Storage location?* App-specific internal/external/SD (Ch 62), not shared.
19. *Cleanup?* Auto-delete watched, expiry, cap, storage pressure.
20. *Per-account?* Filter + isolate downloads/licenses (Ch 42).

Playback

21. *Offline playback?* Same Media3 player reads the download cache + offline license (Ch 58).
22. *Online ↔ offline switch?* Player picks source by availability/preference (transparent).
23. *Completion meaning?* Media fully downloaded AND license stored.

Licensing policy

24. *Limits (titles/devices)?* Server-authorized (Ch 19/23).
25. *Entitlement/geo?* Server-enforced; client UX only.
26. *Can a modified client self-grant?* No — server issues the license.

Security

27. *Protected playback capture?* `FLAG_SECURE`/secure surface (Ch 58).
28. *Network for license/manifest?* Authed + pinned (Ch 19/23).
29. *Logging?* No keys/signed URLs/PII (Ch 28).

Testing/Production

30. *Test completion?* Requires media + license.
31. *Test delete?* Atomic (media+license+row).
32. *Test expiry?* Blocks playback; renewal re-acquires.
33. *Test resume?* After interruption/reboot.
34. *Won't-play in prod?* License not acquired/expired.
35. *Stuck downloads?* Scheduler/constraints/storage.
36. *Orphans?* Non-atomic delete.
37. *Illegal continued play?* License not validated at play.
38. *Over-limit?* Server not enforcing.
39. *Key metrics?* Completion, license-acquire, expiry churn, storage.
40. *Expiry churn?* Titles expiring unwatched — product signal.
41. *What survives reboot?* DownloadManager index + Scheduler resume.

Design-change challenges

1. **Download a DRM movie for offline viewing.** Media3 download stack + offline Widevine license + SSOT + FGS (Ch 46).
2. **A downloaded title must expire after 48h of first play.** Offline license playback window; block + renew (Ch 19).
3. **Deleting a download must free the user's device slot.** Release the license on delete (atomic).
4. **Users on cellular shouldn't auto-download.** Wi-Fi-only constraints (Ch 47).
5. **Play downloads offline, stream otherwise, seamlessly.** `CacheDataSource` + offline license; player picks source (Ch 58).

Mock walkthrough (abridged)

Clarify (DRM?, adaptive?, limits/expiry, storage/SD, cross-device?) → license is the asset → Media3 download stack (`DownloadHelper` track select → `DownloadManager` → `DownloadService` FGS, Ch 46) under constraints (Ch 47) → offline Widevine license acquire/ store/expiry/renew/release (Ch 19/25) → SSOT model + secure/SD storage (Ch 3/25/62) → transparent offline playback (Ch 58) → server-enforced entitlement/limits (Ch 19/23) → cleanup/pressure → observability (completion, license, expiry churn).

59.17 Practical Exercises

1. **Beginner.** Download a chosen-quality track of an adaptive title via Media3 `DownloadService` (FGS) with Wi-Fi-only constraints; model it in the SSOT. *Eval:* downloads in background; state/progress in Room.
2. **Intermediate.** Acquire an offline Widevine license at download time, gate playback on its validity, and release it on delete. *Eval:* plays only with a valid license; delete releases.

3. **Advanced.** Add expiry (playback window) with countdown UI and server-authorized renewal; make deletion atomic (media + license + row). *Eval:* expires/renews; no orphans.
4. **Production debugging.** Fix "downloaded but won't play" (completed before license) and stuck downloads after reboot (no Scheduler). *Eval:* completion requires license; resumes.
5. **System design challenge.** Add storage management (quality estimates, SD choice, auto-delete-watched) and per-account isolation; build a completion/expiry dashboard. *Eval:* storage-sane; isolated; measurable.

(Solutions withheld unless requested.)

59.18 Chapter Summary & Reference Sheet

Key decisions: treat the **offline license as the real asset** (bytes without a valid license are worthless and illegal to play); use **Media3's download stack** — `DownloadHelper` to select a **single chosen rendition** (+ desired audio/subtitle) of an adaptive stream, `DownloadManager` for the resumable queue (persistent index, pause/resume/retry), and a `DownloadService` (foreground service, Ch 46) to run it in the **background** under **constraints** (Wi-Fi-only/charging, Ch 47), surviving reboot/ process death via a **Scheduler** (Ch 30); **acquire/store/renew/release an offline Widevine license** (expiry-bound, keys platform-held, stored securely, Ch 25) where **playback is gated by license validity** and **delete releases the license** atomically with the media and SSOT row; model downloads in the **SSOT** (Ch 3) in **app-specific/SD storage** (Ch 62); play the downloaded copy **transparently** via the same **Media3 player** (`CacheDataSource` + offline license, Ch 58); and **enforce entitlement, device/title limits, geo, and expiry on the server** (Ch 19/23) since a modified client cannot self-grant a license. Downloads/licenses are **per-account** (Ch 42), and storage is actively managed (quality estimates, cleanup, pressure).

Concepts: offline license (acquire/store/expiry/renew/release), Media3 download stack (`DownloadManager` / `DownloadService` / `DownloadHelper`), track/quality selection, FGS background + constraints, SSOT download model, secure/SD storage, transparent offline playback, server-enforced licensing, storage management.

Common mistakes: downloading all bitrates (storage); treating file presence as play permission (DRM/legality); marking "completed" before the license is acquired (won't-play); not releasing the license on delete (over-limit/orphans); not validating license at play (illegal continued play); client-only limit enforcement (bypassable); hand-rolling adaptive+DRM download; no resume/Scheduler (lost on reboot); shared storage; cross-account leakage.

Checklist:

- [ ] Media3 download stack (`DownloadHelper` / `DownloadManager` / `DownloadService` FGS, Ch 46)
- [ ] download one chosen rendition
- [ ] Wi-Fi-only/charging constraints (Ch 47)
- [ ] resume via Scheduler, survive reboot/death (Ch 30)
- [ ] offline Widevine license acquire/store/expiry/renew (Ch 25)
- [ ] playback gated by license validity
- [ ] atomic delete (media + license + SSOT row)
- [ ] SSOT model (Ch 3), app-specific/SD storage (Ch 62)
- [ ] transparent offline playback (Ch 58)
- [ ] server-enforced entitlement/limits/geo/expiry (Ch 19/23)
- [ ] per-account (Ch 42)

- ☐ `FLAG_SECURE` for protected playback
- ☐ storage cleanup/pressure.

Key APIs: Media3 `DownloadManager` / `DownloadService` / `DownloadHelper` / `DownloadRequest` / `Download`, `DownloadNotificationHelper`, `WorkManagerScheduler` / `Requirements` (Ch 30/47), `OfflineLicenseHelper` (`downloadLicense` / `releaseLicense` / `getLicenseDurationRemainingSec`), `MediaItem.DrmConfiguration` / `setKeySetId`, `CacheDataSource` / `Cache`, `C.WIDEVINE_UUID`, `FLAG_SECURE`, Room SSOT (Ch 3), secure storage (Ch 25).

References:

- Media3 downloading: https://developer.android.com/media/media3/exoplayer/downloading-media
- Offline DRM / downloading protected content: https://developer.android.com/media/media3/exoplayer/drm#offline
- DownloadService: https://developer.android.com/reference/androidx/media3/exoplayer/offline/DownloadService
- Store data (app-specific/SD): https://developer.android.com/training/data-storage
- (composes) Ch 58 Media Playback, Ch 18 Downloads, Ch 46 FGS, Ch 30 WorkManager, Ch 47 Constraints, Ch 25 Secure Storage, Ch 62 SAF/Storage, Ch 19/23 Auth/Security

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/flow/failure diagrams; DownloadService + DownloadHelper + OfflineLicenseController + SSOT repo + offline-aware MediaSource code; concurrency (completed=media+license, atomic delete, reboot resume); failure (license-governs-everything); 5 levels; Media3-stack-vs-custom, license-gated, rendition-selection trade-offs; security/DRM (offline license, server auth, FLAG_SECURE, per-account); perf (one rendition, constraints, resume); fakes + instrumentation tests; observability (completion/license/expiry churn); 41 Q&A + 5 challenges. ✅
- **Expand later:** playback (Ch 58), downloads (Ch 18), storage (Ch 62), FGS (Ch 46).
- **Interview coverage:** exceeds ≥40 Complex bar.